



LMR33630 SIMPLE SWITCHER® 3.8-V to 36-V, 3-A Synchronous Step-Down Voltage Converter

1 Features

- Input Voltage Range: 3.8 V to 36 V
- Output DC Current Range: 0 A to 3 A
- Output Voltage Range: 1 V to 24 V
- Integrated Synchronous Rectification
- Low Shutdown Quiescent Current of 6 μ A
- Low Operating Quiescent Current of 25 μ A
- Short Minimum On-time of 70 ns
- High Efficiency at Light Loads
- Peak Current Mode Controls
- Switching Frequency = 400 kHz
- Power-Good Flag for Sequencing, Fault Reporting, and Output Voltage Monitoring
- Precision Enable to Program System UVLO
- Cycle-by-Cycle Current Limiting
- Operating Junction Temperature -40°C to $+125^{\circ}\text{C}$

2 Applications

- Industrial Power Supplies
- Factory and Building Automation Systems
- General-Purpose Wide Input Voltage Regulation

3 Description

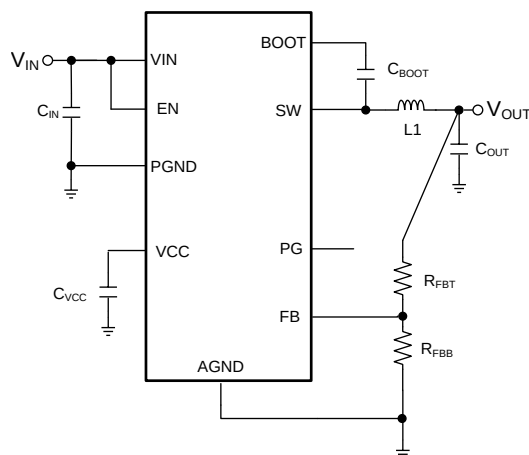
The LMR33630 SIMPLE SWITCHER® regulator is an easy-to-use, synchronous, step-down DC-DC converter capable of driving up to 3 A of load current from an input of up to 36 V. The LMR33630 provides exceptional efficiency and output accuracy in a very small solution size. Peak-current-mode control is employed. Additional features such as a power-good flag and precision enable provide both flexible and easy-to-use solutions for a wide range of applications. The LMR33630 automatically folds back frequency at light load to improve efficiency. The family requires few external components and has a pinout designed for simple PCB layout. Protection features include thermal shutdown, input undervoltage lockout, cycle-by-cycle current limit, and hiccup short-circuit protection. The LMR33630 device is available in an 8-pin HSOIC package.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
LMR33630	HSOIC (8)	6.00 mm $\times$ 5.00 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Schematic



Efficiency vs Output Current $V_{\text{OUT}} = 5 \text{ V}$, 400 kHz, HSOIC

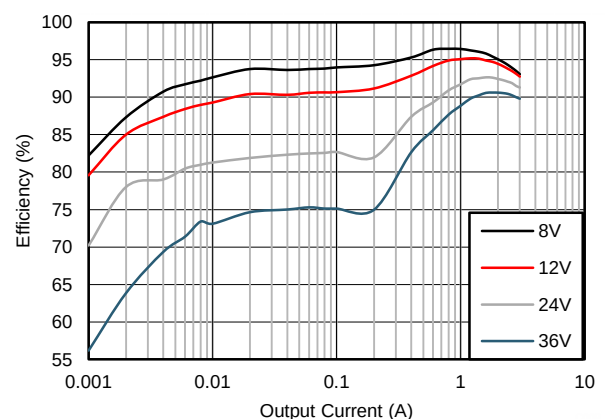


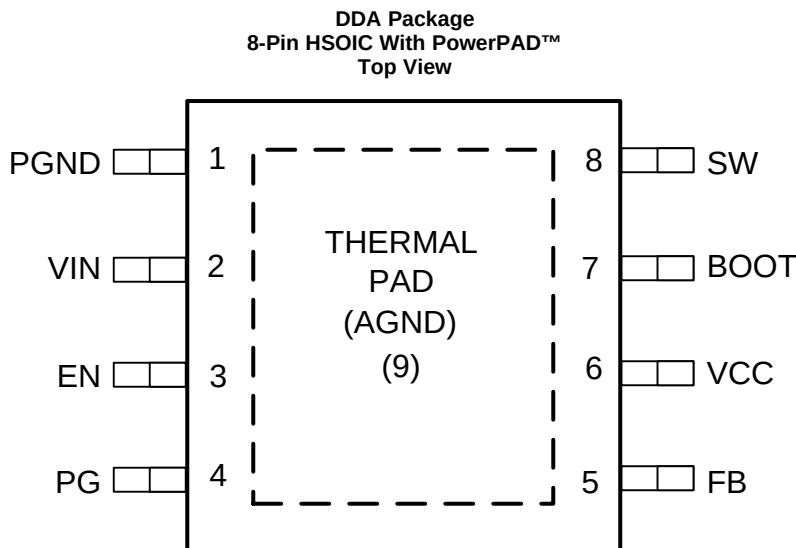
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4 Revision History

DATE	REVISION	NOTES
August 2017	*	Initial release

5 Pin Configuration and Functions



Pin Functions for 8-Pin DDA (HSOIC) Package

NO.	NAME	TYPE	DESCRIPTION
1	PGND	G	Power ground terminal. Connect to system ground and AGND. Connect C_{IN} with short wide traces.
2	VIN	P	Input supply to regulator. Connect a high-quality bypass capacitor(s) directly to this pin and PGND.
3	EN	A	Enable input to regulator. High = ON, Low = OFF. Can be connected directly to VIN; DO NOT FLOAT.
4	PG	A	Open drain power good flag output. Connect to suitable voltage supply through a current limiting resistor. High = power OK, low = power bad. Flag pulls low when EN = Low. Can be open or grounded when not used.
5	FB	A	Feedback input to regulator. Connect to tap point of feedback voltage divider. DO NOT FLOAT. DO NOT GROUND.
6	VCC	P	Internal 5-V LDO output. Used as supply to internal control circuits. Do not connect to external loads. Can be used as logic supply for power-good flag. Connect a high-quality, 1- μ F capacitor from this pin to GND.
7	BOOT	P	Boot-strap supply voltage for internal high-side driver. Connect a high-quality, 100-nF capacitor from this pin to the SW pin.
8	SW	P	Regulator switch node. Connect to power inductor.
9	Thermal Pad (AGND)	G	Analog ground for regulator and system. Ground reference for internal references and logic. All electrical parameters are measured with respect to this pin. Connect to system ground on PCB.

A = Analog, P = Power, G = Ground

6 Specifications

6.1 Absolute Maximum Ratings

Over the recommended operating junction temperature range⁽¹⁾

PARAMETER		MIN	MAX	UNIT
Voltages	VIN to PGND	−0.3	38	V
	EN to AGND ⁽²⁾	−0.3	V _{IN} + 0.3	
	FB to AGND	−0.3	5.5	
	PG to AGND	0	22	
	AGND to PGND ⁽²⁾	−0.3	0.3	
	SW to PGND	−0.3	V _{IN} + 0.3	V
	SW to PGND less than 100-ns transients	−3.5	38	
	BOOT to SW	−0.3	5.5	
	VCC to AGND	−0.3	6	
Current	PG to AGND ⁽³⁾	0	4	mA
T _J	Junction temperature ⁽⁴⁾	−40	150	°C
T _{stg}	Storage temperature	−65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) The voltage on this pin must not exceed the voltage on the VIN pin by more than 0.3 V
- (3) Positive current flows into this pin.
- (4) Operating at junction temperatures greater than 125°C, although possible, degrades the lifetime of the device.

6.2 ESD Ratings

		VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM) ⁽¹⁾	±1000
		Charged-device model (CDM) ⁽²⁾	±500

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

Over the recommended operating junction temperature range of −40°C to +125°C (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
Input voltage	VIN to PGND	3.8	36	V
	EN ⁽²⁾	0	V _{IN}	
	PG ⁽²⁾	0	12	
Output voltage	V _{OUT} ⁽³⁾	1	24	V
Output current	I _{OUT}	0	3	A
Temperature	Operating junction temperature, T _J ⁽⁴⁾	−40	125	°C

- (1) Recommended operating conditions indicate conditions for which the device is intended to be functional, but do not ensure specific performance limits. For ensured specifications, see *Electrical Characteristics*.
- (2) The voltage on this pin must not exceed the voltage on the VIN pin by more than 0.3 V.
- (3) Under no conditions should the output voltage be allowed to fall below zero volts.
- (4) Operating at junction temperatures greater than 125°C, although possible, degrades the lifetime of the device.

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾ (2)		LMR33630	UNIT
		DDA (HSOIC)	
		8 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	42.9	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	54	°C/W
R _{θJB}	Junction-to-board thermal resistance	13.6	°C/W
ψ _{JT}	Junction-to-top characterization parameter	4.3	°C/W
ψ _{JB}	Junction-to-board characterization parameter	13.8	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	4.3	°C/W

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.
- (2) The value of R_{θJA} given in this table is only valid for comparison with other packages and can not be used for design purposes. These values were calculated in accordance with JESD 51-7, and simulated on a 4-layer JEDEC board. They do not represent the performance obtained in an actual application. For design information, see [Maximum Ambient Temperature](#).

6.5 Electrical Characteristics

Limits apply over the recommended operating junction temperature (T_J) range of –40°C to +125°C, unless otherwise stated. Minimum and maximum limits are specified through test, design or statistical correlation. Typical values represent the most likely parametric norm at T_J = 25°C, and are provided for reference purposes only. Unless otherwise stated, the following conditions apply: V_{IN} = V_{EN} = 12 V.

PARAMETER		TEST CONDITIONS	MIN ⁽¹⁾	TYP	MAX ⁽¹⁾	UNIT
SUPPLY VOLTAGE						
V _{IN}	Operating input voltage range		3.8		36	V
I _Q	Non-switching input current; measured at VIN pin ⁽²⁾	V _{FB} = 1.2 V		24	29	μA
I _{SD}	Shutdown quiescent current; measured at VIN pin	EN = 0		6	10	μA
ENABLE						
V _{EN-VCC-H}	EN input level required to turn on internal LDO	Rising threshold			1	V
V _{EN-VCC-L}	EN input level required to turn off internal LDO	Falling threshold	0.3			V
V _{EN-H}	EN input level required to start switching	Rising threshold	1.2	1.231	1.26	V
V _{EN-HYS}	Hysteresis below V _{EN-H}	Hysteresis below V _{EN-H} ; falling		100		mV
I _{LKG-EN}	Enable input leakage current	V _{EN} = 3.3 V		0.2		nA
INTERNAL SUPPLIES						
V _{CC}	Internal LDO output voltage appearing at the VCC pin	6 V ≤ V _{IN} ≤ 36 V	4.75	5	5.25	V
V _{BOOT-UVLO}	Bootstrap voltage undervoltage lock-out threshold ⁽³⁾			2.6		V
VOLTAGE REFERENCE (FB PIN)						
V _{FB}	Feedback voltage		0.985	1	1.015	V
I _{LKG-FB}	Leakage current into FB pin	FB = 1 V		0.2	10	nA

- (1) MIN and MAX limits are 100% production tested at 25°C. Limits over the operating temperature range verified through correlation using Statistical Quality Control (SQC) methods. Limits are used to calculate Average Outgoing Quality Level (AOQL).
- (2) This is the current used by the device open loop. It does not represent the total input current of the system when in regulation.
- (3) When the voltage across the C_{BOOT} capacitor falls below this voltage, the low side MOSFET is turned on to recharge C_{BOOT}.

Electrical Characteristics (continued)

Limits apply over the recommended operating junction temperature (T_J) range of -40°C to $+125^{\circ}\text{C}$, unless otherwise stated. Minimum and maximum limits are specified through test, design or statistical correlation. Typical values represent the most likely parametric norm at $T_J = 25^{\circ}\text{C}$, and are provided for reference purposes only. Unless otherwise stated, the following conditions apply: $V_{IN} = V_{EN} = 12\text{ V}$.

PARAMETER		TEST CONDITIONS	MIN ⁽¹⁾	TYP	MAX ⁽¹⁾	UNIT
CURRENT LIMITS⁽⁴⁾						
I_{SC}	High-side current limit	LMR33630	3.8	4.5	5.2	A
I_{LIMIT}	Low-side current limit	LMR33630	3	3.5	4	A
$I_{PEAK-MIN}$	Minimum peak inductor current	LMR33630		0.8		A
I_{ZC}	Zero current detector threshold			0.01		A
SOFT START						
t_{SS}	Internal soft-start time		3	4	6	ms
POWER GOOD (PG PIN)						
$V_{PG-HIGH-UP}$	Power-good upper threshold - rising	% of FB voltage	105%	107%	110%	
$V_{PG-HIGH-DN}$	Power-good upper threshold - falling	% of FB voltage	103%	105%	108%	
$V_{PG-LOW-UP}$	Power-good lower threshold - rising	% of FB voltage	92%	95%	97%	
$V_{PG-LOW-DN}$	Power-good lower threshold - falling	% of FB voltage	90%	93%	95%	
t_{PG}	Power-good glitch filter delay ⁽⁵⁾		80		170	μs
R_{PG}	Power-good flag R_{DS-ON}	$V_{IN} = 2\text{ V}, V_{EN} = 0\text{ V}$		50		Ω
		$V_{IN} = V_{EN} = 12\text{ V}$		40	125	
		$V_{EN} = 0\text{ V}$		60	150	
V_{PG}	Minimum input voltage for proper PG function	50- μA pullup current, $EN = 0\text{ V}, T_J = 25^{\circ}\text{C}$			2	V
OSCILLATOR						
f_{SW}	Switching frequency in CCM	LMR33630A	340	400	460	kHz
MOSFETS						
$R_{DS-ON-HS}$	High-side MOSFET ON-resistance	DDA package		95		
$R_{DS-ON-LS}$	Low-side MOSFET ON-resistance			75		

(4) The current limit values in this table are tested, open loop, in production. They may differ from those found in a closed loop application.

(5) See [Power-Good Flag Output](#) for details.

6.6 Timing Characteristics

Limits apply over the recommended operating junction temperature (T_J) range of -40°C to $+125^{\circ}\text{C}$, unless otherwise stated. Minimum and maximum limits are specified through test, design or statistical correlation. Typical values represent the most likely parametric norm at $T_J = 25^{\circ}\text{C}$, and are provided for reference purposes only. Unless otherwise stated, the following conditions apply: $V_{IN} = V_{EN} = 12\text{ V}$.

			MIN	NOM	MAX	UNIT
t_{ON-MIN}	Minimum switch on-time	DDA package		70	100	ns
$t_{OFF-MIN}$	Minimum switch off-time	DDA package		50	80	ns
t_{ON-MAX}	Maximum switch on-time			6.2		μs

6.7 System Characteristics

The following specifications apply only to the typical applications circuit shown in [Figure 34](#), with nominal component values. Specifications in the typical (TYP) column apply to $T_J = 25^\circ\text{C}$ only. Specifications in the minimum (MIN) and maximum (MAX) columns apply to the case of typical components over the temperature range of $T_J = -40^\circ\text{C}$ to 125°C . *These specifications are not ensured by production testing.*

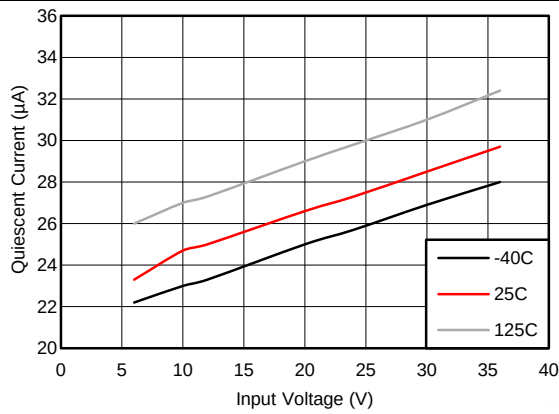
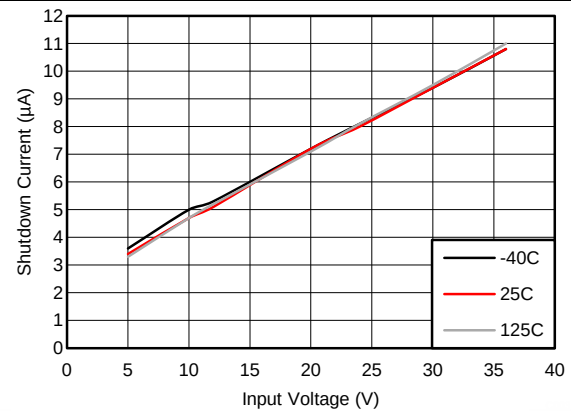
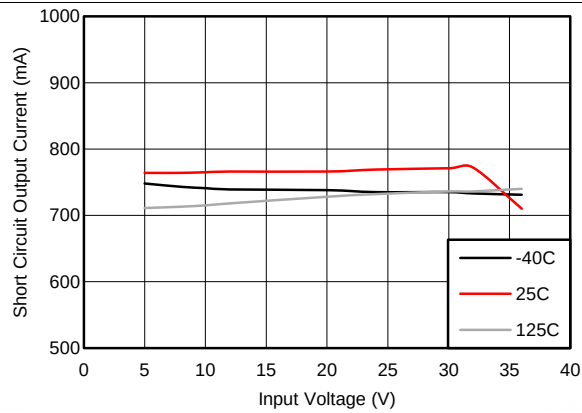
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{IN}	Operating input voltage range	$V_{OUT} = 3.3\text{ V}$, $I_{OUT} = 0\text{ A}$	3.8		36	V
V_{OUT}	Output voltage regulation for $V_{OUT} = 5\text{ V}^{(1)}$	$V_{OUT} = 5\text{ V}$, $V_{IN} = 7\text{ V}$ to 36 V , $I_{OUT} = 0\text{ A}$ to 3 A	–1.5%		2.5%	
		$V_{OUT} = 5\text{ V}$, $V_{IN} = 7\text{ V}$ to 36 V , $I_{OUT} = 1\text{ A}$ to 3 A	–1.5%		1.5%	
	Output voltage regulation for $V_{OUT} = 3.3\text{ V}^{(1)}$	$V_{OUT} = 3.3\text{ V}$, $V_{IN} = 3.8\text{ V}$ to 36 V , $I_{OUT} = 0\text{ A}$ to 3 A	–1.5%		2.5%	
		$V_{OUT} = 3.3\text{ V}$, $V_{IN} = 3.8\text{ V}$ to 36 V , $I_{OUT} = 1\text{ A}$ to 3 A	–1.5%		1.5%	
I_{SUPPLY}	Input supply current when in regulation	$V_{IN} = 12\text{ V}$, $V_{OUT} = 3.3\text{ V}$, $I_{OUT} = 0\text{ A}$, $R_{FBT} = 1\text{ M}\Omega$		25		μA
V_{DROP}	Dropout voltage; ($V_{IN} - V_{OUT}$)	$V_{OUT} = 5\text{ V}$ Dropout at –1% of regulation, $f_{SW} = 140\text{ kHz}$		150		mV
D_{MAX}	Maximum switch duty cycle ⁽²⁾	$V_{IN} = V_{OUT} = 12\text{ V}$, $I_{OUT} = 1\text{ A}$		98%		
V_{HC}	FB pin voltage required to trip short-circuit hiccup mode			0.4		V
t_{HC}	Time between current-limit hiccup burst			94		ms
t_D	Switch voltage dead time			2		ns
T_{SD}	Thermal shutdown temperature	Shutdown temperature		165		$^\circ\text{C}$
		Recovery temperature		148		$^\circ\text{C}$

(1) Deviation is with respect to $V_{IN} = 12\text{ V}$, $I_{OUT} = 1\text{ A}$.

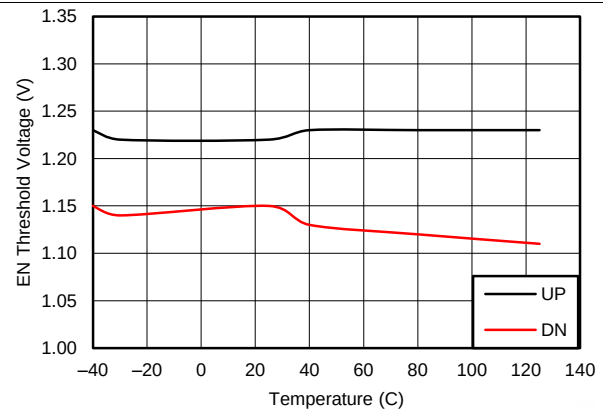
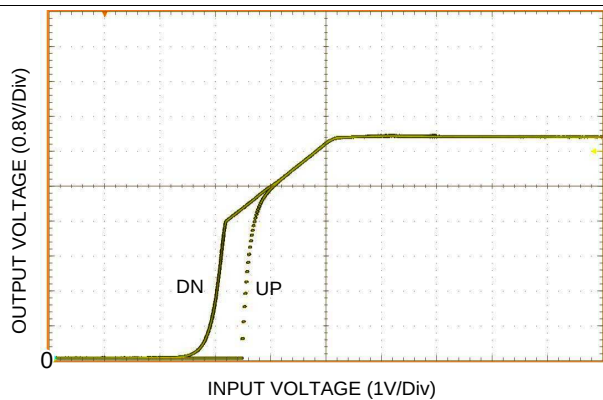
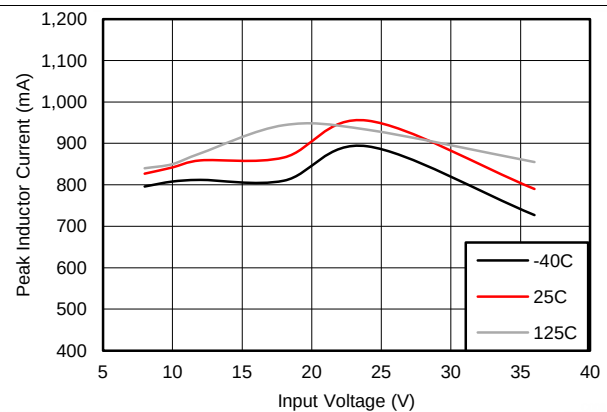
(2) In dropout the switching frequency drops to increase the effective duty cycle. The lowest frequency is clamped at approximately: $f_{MIN} = 1 / (t_{ON-MAX} + t_{OFF-MIN})$. $D_{MAX} = (t_{ON-MAX} + t_{OFF-MIN})$.

6.8 Typical Characteristics

Unless otherwise specified the following conditions apply: $T_A = 25^\circ\text{C}$. $V_{IN} = 12\text{ V}$


 $V_{FB} = 1.2\text{ V}$
Figure 1. Non-Switching Input Supply Current

 $EN = 0\text{ V}$
Figure 2. Shutdown Supply Current

 $V_{OUT} = 0\text{ V}$
 $f_S = 400\text{ kHz}$

See Figure 34

Figure 3. Short Circuit Output Current

Figure 4. Precision Enable Thresholds

 $I_{OUT} = 1\text{ mA}$
Figure 5. UVLO Thresholds

 $I_{OUT} = 0\text{ A}$
 $f_{SW} = 400\text{ kHz}$
 $V_{OUT} = 5\text{ V}$

See Figure 34

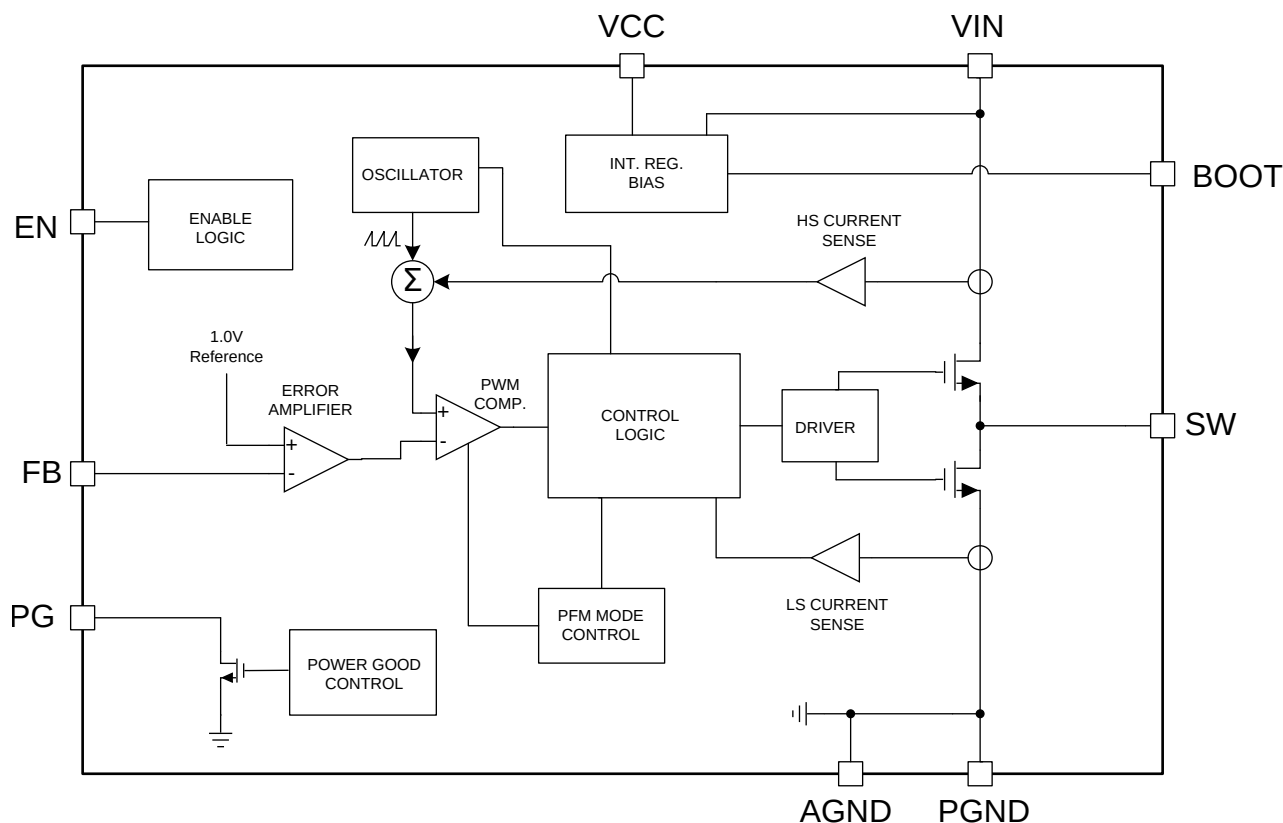
Figure 6. $I_{PEAK-MIN}$

7 Detailed Description

7.1 Overview

The LMR33630 is a synchronous peak-current-mode buck regulator designed for a wide variety of industrial applications. The regulator automatically switches modes between PFM and PWM depending on load. At heavy load, the device operates in PWM at a constant switching frequency. At light loads the mode changes to PFM, with diode emulation allowing DCM. This reduces the input supply current and keeps efficiency high. The device features internal loop compensation which reduces design time and requires fewer external components than externally compensated regulators.

7.2 Functional Block Diagram



ADVANCE INFORMATION

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7.3 Feature Description

7.3.1 Power-Good Flag Output

The power-good flag function (PG output pin) of the LMR33630 can be used to reset a system microprocessor whenever the output voltage is out of regulation. This open-drain output goes low under fault conditions, such as current limit and thermal shutdown, as well as during normal start-up. A glitch filter prevents false flag operation for short excursions of the output voltage, such as during line and load transients. The timing parameters of the glitch filter are found in the [Electrical Characteristics](#) table. Output voltage excursions lasting less than T_{PG} do not trip the power-good flag. Power-good operation can best be understood by reference to [Figure 7](#) and [Figure 8](#). Note that during initial power-up a delay of about 4 ms (typical) is inserted from the time that EN is asserted to the time that the power-good flag goes high. This delay only occurs during start-up and is not encountered during normal operation of the power-good function.

Feature Description (continued)

The power-good output consists of an open drain NMOS; requiring an external pull up resistor to a suitable logic supply. It can also be pulled up to either V_{CC} or V_{OUT} , through an appropriate resistor, as desired. If this function is not needed, the PG pin should be grounded. When EN is pulled low, the flag output is also forced low. With EN low, power good remains valid as long as the input voltage is ≥ 2 V (typical). Limit the current into this pin to ≤ 4 mA.

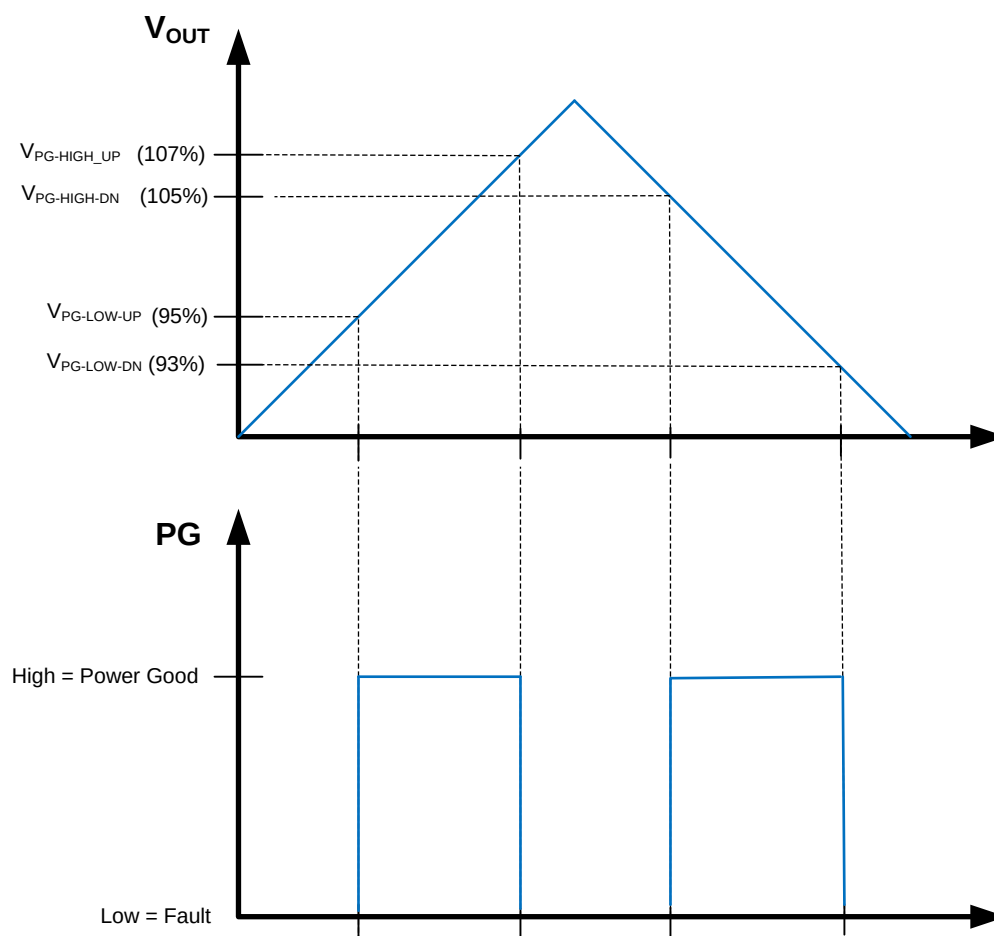


Figure 7. Static Power-Good Operation

Feature Description (continued)

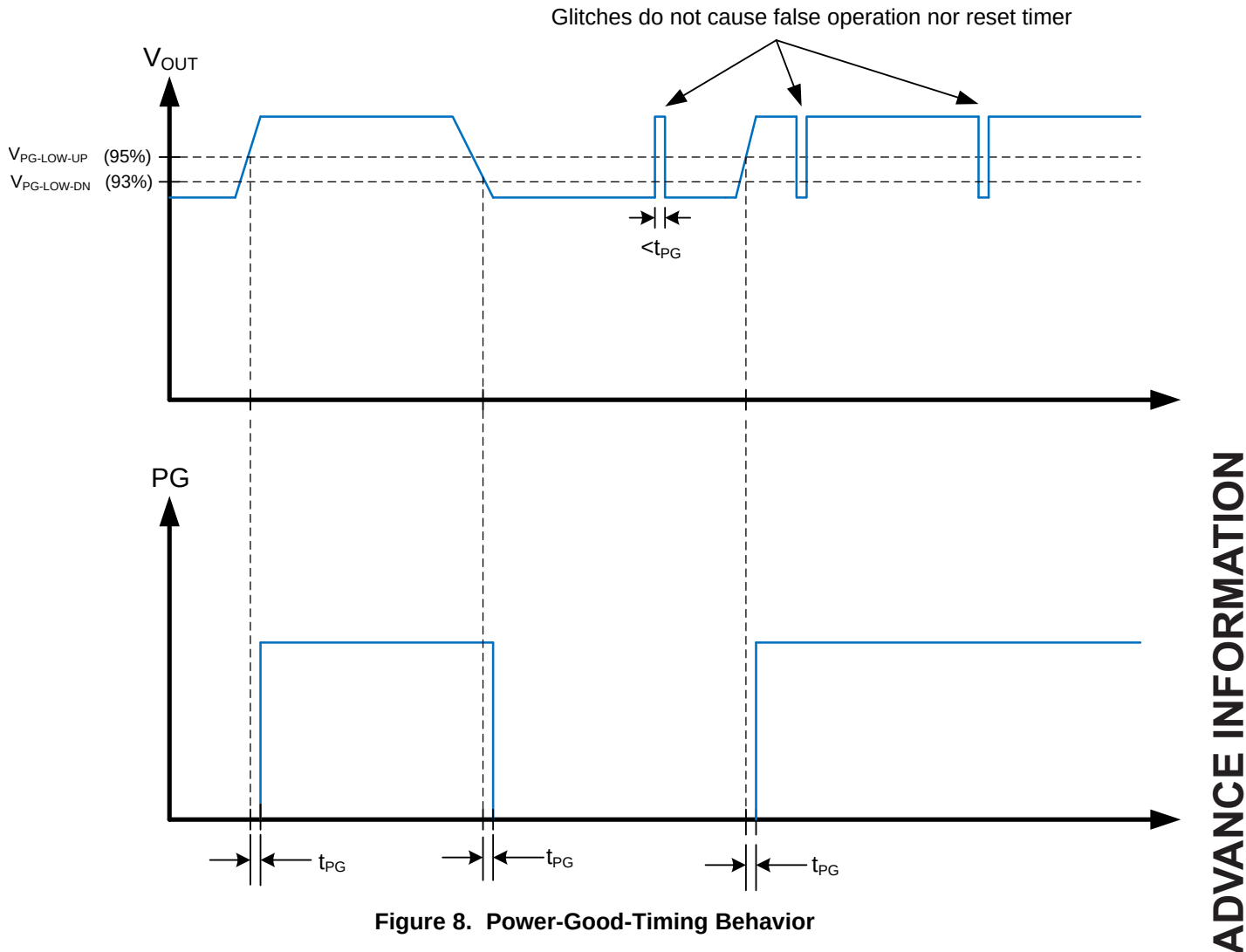


Figure 8. Power-Good-Timing Behavior

7.3.2 Enable and Start-up

Start-up and shutdown are controlled by the EN input. This input features precision thresholds, allowing the use of an external voltage divider to provide an adjustable input UVLO (see the [External UVLO](#) section). Applying a voltage of $\geq V_{EN-VCC_H}$ causes the device to enter standby mode, powering the internal VCC, but not producing an output voltage. Increasing the EN voltage to V_{EN-H} fully enables the device, allowing it to enter start-up mode and beginning the soft-start period. When the EN input is brought below V_{EN-H} by V_{EN-HYS} , the regulator stops running and enters standby mode. Further decrease in the EN voltage to below $V_{EN-VCC-L}$ completely shuts down the device. This behavior is shown in [Figure 9](#). The EN input may be connected directly to VIN if this feature is not needed. This input must not be allowed to float. The values for the various EN thresholds can be found in the [Electrical Characteristics](#).

The LMR33630 utilizes a reference-based soft start that prevents output voltage overshoots and large inrush currents as the regulator is starting up. A typical start-up waveform is shown in [Figure 10](#) along with typical timings. The rise time of the output voltage is about 4 ms (see [Electrical Characteristics](#)).

Feature Description (continued)

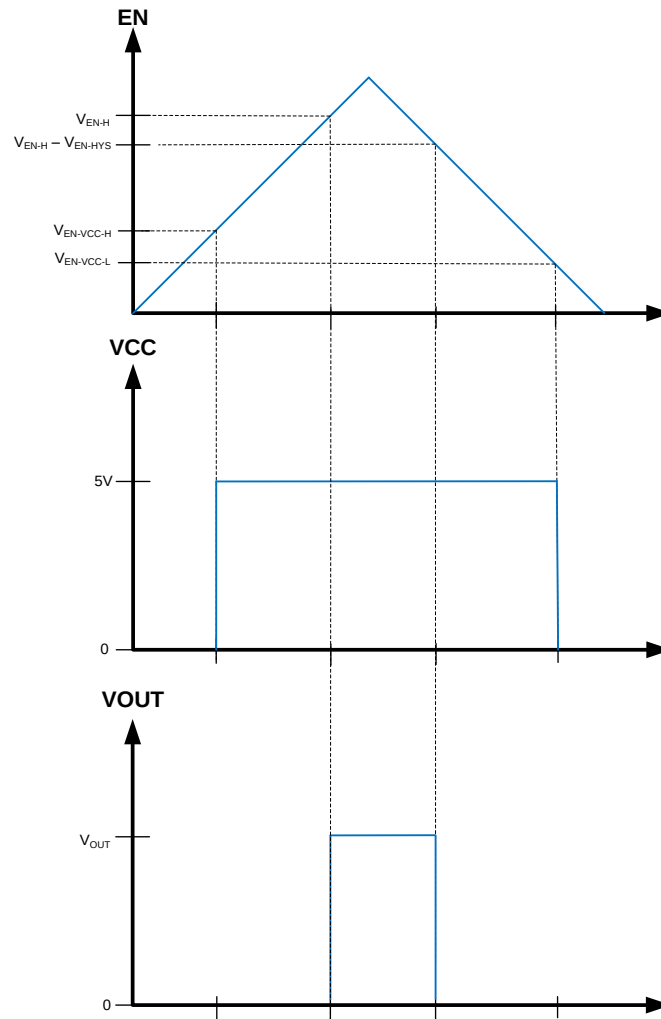


Figure 9. Precision Enable Behavior

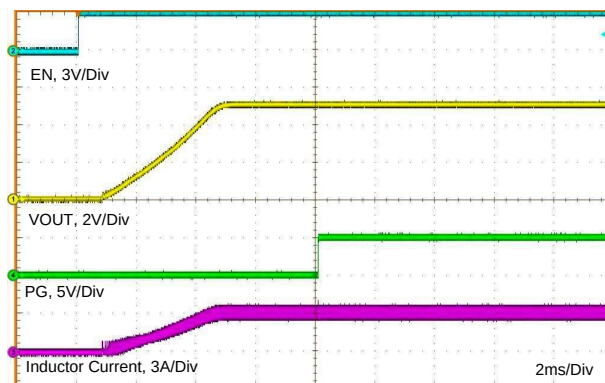


Figure 10. Typical Start-up Behavior
 $V_{IN} = 12\text{ V}$, $V_{OUT} = 5\text{ V}$, $I_{OUT} = 3\text{ A}$

Feature Description (continued)

7.3.3 Current Limit and Short Circuit

The LMR33630 incorporates valley current limit for normal overloads and for short-circuit protection. In addition the high-side power MOSFET is protected from excessive current by a peak current limit circuit. Cycle-by-cycle current limit is used for overloads, while hiccup mode is used for short circuits. Finally, a zero current detector is used on the low-side power MOSFET to implement DEM at light loads (see [Glossary](#)). The typical value of this current limit is found under I_{ZC} in the [Electrical Characteristics](#).

During overloads the low-side current limit, I_{LIMIT} , (see [Electrical Characteristics](#) table) determines the maximum load current that the LMR33630 can supply. When the low-side switch turns on, the inductor current begins to ramp down. If the current does not fall below I_{LIMIT} before the next turn-on cycle, then that cycle is skipped, and the low-side MOSFET is left on until the current falls below I_{LIMIT} . This is somewhat different than the more typical peak current limit and results in [Equation 1](#) for the maximum load current.

$$I_{OUT}|_{max} = I_{LIMIT} + \frac{(V_{IN} - V_{OUT})}{2 \cdot f_{SW} \cdot L} \cdot \frac{V_{OUT}}{V_{IN}}$$

where

- f_{SW} = Switching frequency
- L = Inductor value

(1)

If, during current limit, the voltage on the FB input falls below about 0.4 V, due to a short circuit, the device enters into hiccup mode. In this mode the device stops switching for t_{HC} (see [System Characteristics](#)), or about 94 ms and then goes through a normal re-start with soft start. If the short-circuit condition remains, the device runs in current limit for about 20 ms (typical) and then shuts down again. This cycle repeats, as shown in [Figure 11](#) as long as the short-circuit-condition persists. This mode of operation helps to reduce the temperature rise of the device during a hard short on the output. Of course the output current is greatly reduced during hiccup mode (see [Typical Characteristics](#)). Once the output short is removed, and the hiccup delay is passed, the output voltage recovers normally as shown in [Figure 12](#)

The high-side-current limit trips when the peak inductor current reaches I_{SC} (see [Electrical Characteristics](#) table). This is a cycle-by-cycle current limit and does not produce any frequency or load current fold back. It is meant to protect the high-side MOSFET from excessive current. Under some conditions, such as high input voltages, this current limit may trip before the low-side protection. I_{SC} varies with duty cycle.



Figure 11. Inductor Current Burst in Short-Circuit Mode

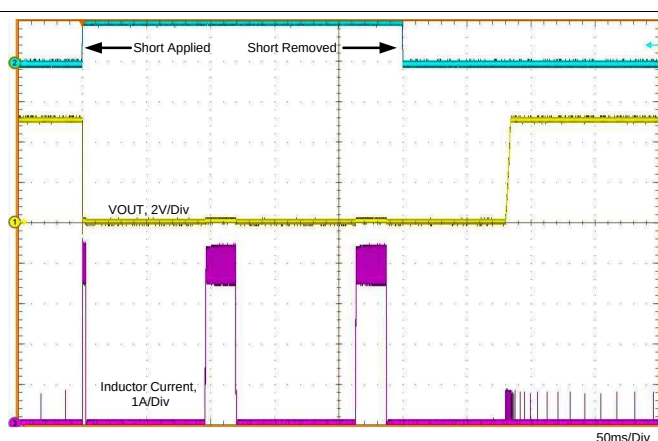


Figure 12. Short-Circuit Transient and Recovery

Feature Description (continued)

7.3.4 Undervoltage Lockout and Thermal Shutdown

The LMR33630 incorporates an undervoltage-lockout feature on the output of the internal LDO (at the VCC pin). When VCC reaches about 3.7 V the device is ready to receive an EN signal and start up. When VCC falls below about 3 V the device shuts down, regardless of EN status. Since the LDO is in dropout during these transitions, the above values roughly represent the input voltage levels during the transitions.

Thermal shutdown is provided to protect the regulator from excessive junction temperature. When the junction temperature reaches about 165°C the device shuts down; re-start occurs when the temperature falls to about 148°C .

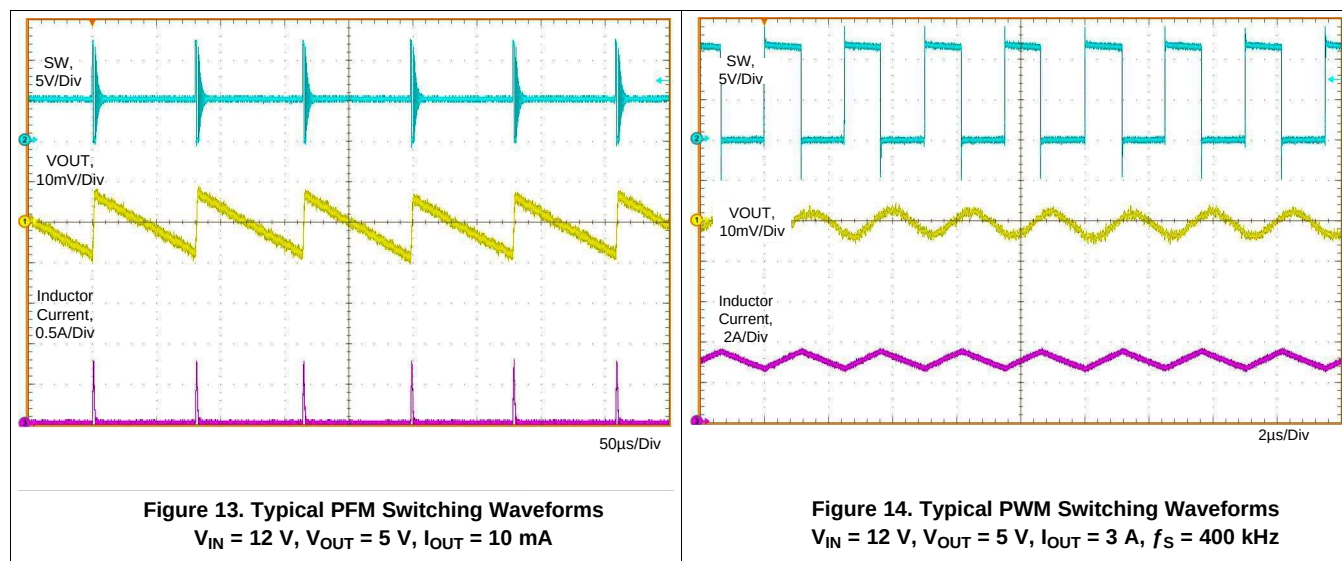
7.4 Device Functional Modes

7.4.1 Auto Mode

In auto mode the device moves between PWM and PFM as the load changes. At light loads the regulator operates in PFM. At higher loads the mode changes to PWM. The load current for which the device moves from PFM to PWM can be found in the [Application Curves](#).

In PWM the regulator operates as a constant frequency, current mode, full synchronous converter using PWM to regulate the output voltage. While operating in this mode the output voltage is regulated by switching at a constant frequency and modulating the duty cycle to control the power to the load. This provides excellent line and load regulation and low output voltage ripple.

In PFM the high-side MOSFET is turned on in a burst of one or more pulses to provide energy to the load. The duration of the burst depends on how long it takes the inductor current to reach $I_{PEAK-MIN}$. The frequency of these bursts is adjusted to regulate the output, while diode emulation (DEM) is used to maximize efficiency (see [Glossary](#)). This mode provides high light-load efficiency by reducing the amount of input supply current required to regulate the output voltage at small loads. This trades off very good light-load efficiency for larger output voltage ripple and variable switching frequency. Also, a small increase in output voltage occurs at light loads. The actual switching frequency and output voltage ripple depends on the input voltage, output voltage, and load. Typical switching waveforms in PFM and PWM are shown in [Figure 13](#) and [Figure 14](#). See the [Application Curves](#) for output voltage variation with load in auto mode.



Device Functional Modes (continued)

7.4.2 Dropout

The dropout performance of any buck regulator is affected by the $R_{DS(on)}$ of the power MOSFETs, the DC resistance of the inductor, and the maximum duty cycle that the controller can achieve. As the input voltage is reduced to near the output voltage, the off-time of the high side MOSFET starts to approach the minimum value (see [Timing Characteristics](#)). Beyond this point the switching may become erratic and/or the output voltage will fall out of regulation. To avoid this problem the LMR33630 automatically reduces the switching frequency to increase the effective duty cycle and maintain regulation. In this data sheet the drop-out voltage is defined as the difference between the input and output voltage when the output has dropped by 1% of its nominal value. Under this condition the switching frequency has dropped to its minimum value of about 140 kHz. See [Application Curves](#) for typical values of dropout voltage under various conditions. Note that the 0.4 V short circuit detection threshold is not activated when in dropout mode. The overall dropout characteristic can be seen in [Figure 15](#).

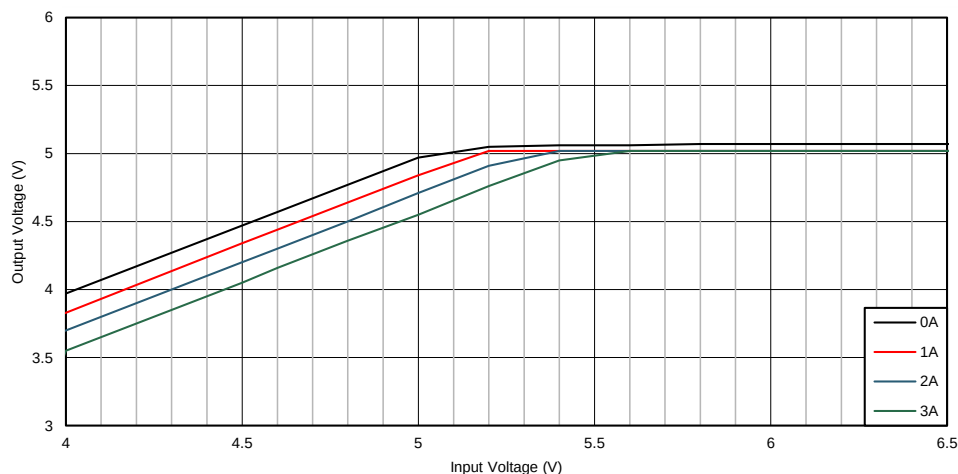


Figure 15. Overall Dropout Characteristic
V_{OUT} = 5 V

7.4.3 Minimum Switch On-time

Every switching regulator has a minimum controllable on-time dictated by the inherent delays and blanking times associated with the control circuits. This imposes a minimum switch duty cycle and therefore a minimum conversion ratio. The constraint is encountered at high input voltages and low output voltages. To help extend the minimum controllable duty cycle, the LMR33630 automatically reduces the switching frequency when the minimum on-time limit is reached. In this way the converter can regulate the lowest programmable output voltage at the maximum input voltage. An estimate for the approximate input voltage, for a given output voltage, before frequency fold-back occurs is found in [Equation 2](#). The values of t_{ON} and f_{SW} can be found in the [Electrical Characteristics](#) table. As the input voltage is increased, the switch on-time (duty-cycle) reduces to regulate the output voltage. When the on-time reaches the limit, the switching frequency drops, while the on-time remains fixed.

$$V_{IN} \leq \frac{V_{OUT}}{t_{ON} \cdot f_{SW}} \quad (2)$$

8 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

NOTE

All of the capacitance values given in the following application information refer to *effective* values; unless otherwise stated. The *effective* value is defined as the actual capacitance under D.C. bias and temperature; not the rated or nameplate values. Use high-quality, low-ESR, ceramic capacitors with an X5R or better dielectric throughout. All high value ceramic capacitors have a large voltage coefficient in addition to normal tolerances and temperature effects. Under D.C. bias the capacitance drops considerably. Large case sizes and/or higher voltage ratings are better in this regard. To help mitigate these effects, multiple capacitors can be used in parallel to bring the minimum *effective* capacitance up to the required value. This can also ease the RMS current requirements on a single capacitor. A careful study of bias and temperature variation of any capacitor bank should be made in order to ensure that the minimum value of *effective* capacitance is provided.

8.1 Application Information

The LMR33630 step-down DC-to-DC converter is typically used to convert a higher DC voltage to a lower DC voltage with a maximum output current of 3 A. The following design procedure can be used to select components for the LMR33630. Alternately, the WEBENCH® Design Tool may be used to generate a complete design. This tool utilizes an iterative design procedure and has access to a comprehensive database of components. This allows the tool to create an optimized design and allows the user to experiment with various options.

8.2 Typical Application

Figure 16 shows a typical application circuit for the LMR33630. This device is designed to function over a wide range of external components and system parameters. However, the internal compensation is optimized for a certain range of external inductance and output capacitance. As a quick start guide, Table 2 provides typical component values for a range of the most common output voltages.

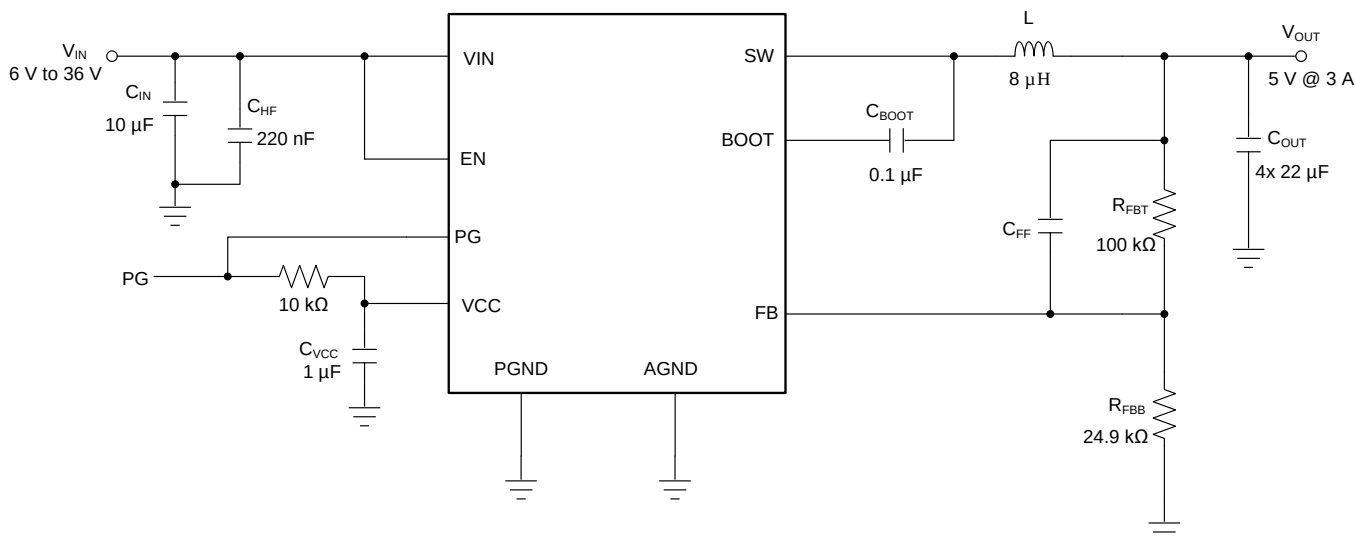


Figure 16. Example Application Circuit (400kHz)

Typical Application (continued)

8.2.1 Design Requirements

The following provides a detailed design procedure based on [Table 1](#).

Table 1. Detailed Design Parameters

DESIGN PARAMETER	EXAMPLE VALUE
Input voltage	12 V (6 V to 36 V)
Output voltage	5 V
Maximum output current	0 A to 3 A
Switching frequency	400 kHz

Table 2. Typical External Component Values

f_{sw} (kHz)	V_{OUT} (V)	L (μ H)	C_{OUT} (rated capacitance)	R_{FBT} (Ω)	R_{FBB} (Ω)	C_{IN}	C_{BOOT}	C_{VCC}	C_{FF}
400	3.3	6.8	$4 \times 22 \mu F$	100 k	43.2 k	$10 \mu F + 220 nF$	100 nF	1 μF	open
400	5	8	$4 \times 22 \mu F$	100 k	24.9 k	$10 \mu F + 220 nF$	100 nF	1 μF	open
400	12	15	$4 \times 10 \mu F$	100 k	9.09 k	$10 \mu F + 220 nF$	100 nF	1 μF	open

8.2.2 Detailed Design Procedure

The following design procedure applies to [Figure 16](#) and [Table 1](#).

8.2.2.1 Choosing the Switching Frequency

The choice of switching frequency is a compromise between conversion efficiency and overall solution size. Lower switching frequency implies reduced switching losses and usually results in higher system efficiency. However, higher switching frequency allows the use of smaller inductors and output capacitors, and hence a more compact design.

8.2.2.2 Setting the Output Voltage

The output voltage of LMR33630 is externally adjustable using a resistor divider network. The range of recommended output voltage is found in the [Recommended Operating Conditions](#) table. The divider network is comprised of R_{FBT} and R_{FBB} , and closes the loop between the output voltage and the converter. The converter regulates the output voltage by holding the voltage on the FB pin equal to the internal reference voltage, V_{REF} . The resistance of the divider is a compromise between excessive noise pick-up and excessive loading of the output. Smaller values of resistance reduce noise sensitivity but also reduce the light-load efficiency. The recommended value for R_{FBT} is 100 k Ω ; with a maximum value of 1 M Ω . If a 1 M Ω is selected for R_{FBT} , then a feed-forward capacitor must be used across this resistor to provide adequate loop phase margin (see [\$C_{FF}\$ Selection](#)). Once R_{FBT} is selected, [Equation 3](#) is used to select R_{FBB} . V_{REF} is nominally 1 V (see [Electrical Characteristics](#) for limits).

$$R_{FBB} = \frac{R_{FBT}}{\left[\frac{V_{OUT}}{V_{REF}} - 1 \right]} \quad (3)$$

For this 5-V example we choose: $R_{FBT} = 100 \text{ k}\Omega$ and $R_{FBB} = 24.9 \text{ k}\Omega$.

8.2.2.3 Inductor Selection

The parameters for selecting the inductor are the inductance and saturation current. The inductance is based on the desired peak-to-peak ripple current and is normally chosen to be in the range of 20% to 40% of the maximum output current. Experience shows that the best value for inductor ripple current is 30% of the maximum load current. Note that when selecting the ripple current for applications with much smaller maximum load than the maximum available from the device, the maximum device current should be used. [Equation 4](#) can be used to determine the value of inductance. The constant K is the percentage of inductor current ripple. For this example we choose K = 0.3 and find an inductance L = 8.1 μ H; we select the next standard value of 8 μ H.

$$L = \frac{(V_{IN} - V_{OUT})}{f_{SW} \cdot K \cdot I_{OUTmax}} \cdot \frac{V_{OUT}}{V_{IN}} \quad (4)$$

Ideally, the saturation current rating of the inductor should be at least as large as the high-side switch current limit, I_{SC} (see [Electrical Characteristics](#)). This ensures that the inductor does not saturate even during a short circuit on the output. When the inductor core material saturates, the inductance falls to a very low value, causing the inductor current to rise very rapidly. Although the valley current limit, I_{LIMIT} , is designed to reduce the risk of current run-away, a saturated inductor can cause the current to rise to high values very rapidly. This may lead to component damage; do not allow the inductor to saturate! Inductors with a ferrite core material have very *hard* saturation characteristics, but usually have lower core losses than powdered iron cores. Powdered iron cores exhibit a *soft* saturation, allowing some relaxation in the current rating of the inductor. However, they have more core losses at frequencies above about 1 MHz. In any case the inductor saturation current should not be less than the device low-side current limit, I_{LIMIT} (see [Electrical Characteristics](#)).

8.2.2.4 Output Capacitor Selection

The value of the output capacitor, and its ESR, determine the output voltage ripple and load transient performance. The output capacitor bank is usually limited by the load transient requirements, rather than the output voltage ripple. [Equation 5](#) can be used to estimate a lower bound on the total output capacitance, and an upper bound on the ESR, required to meet a specified load transient.

$$C_{OUT} \geq \frac{\Delta I_{OUT}}{f_{SW} \cdot \Delta V_{OUT} \cdot K} \cdot \left[(1-D) \cdot (1+K) + \frac{K^2}{12} \cdot (2-D) \right]$$

$$ESR \leq \frac{(2+K) \cdot \Delta V_{OUT}}{2 \cdot \Delta I_{OUT} \left[1+K + \frac{K^2}{12} \cdot \left(1 + \frac{1}{(1-D)} \right) \right]}$$

$$D = \frac{V_{OUT}}{V_{IN}}$$

where

- ΔV_{OUT} = output voltage transient
- ΔI_{OUT} = output current transient
- K = Ripple factor from [Inductor Selection](#)

(5)

Once the output capacitor and ESR have been calculated, [Equation 6](#) can be used to check the output voltage ripple. The output capacitor and ESR can then be adjusted to meet both the load transient and output ripple requirements.

$$V_r \cong \Delta I_L \cdot \sqrt{ESR^2 + \frac{1}{(8 \cdot f_{SW} \cdot C_{OUT})^2}}$$

where

- V_r = Peak-to-peak output voltage ripple

(6)

For this example we require a ΔV_{OUT} of ≤ 250 mV for an output current step of $\Delta I_{OUT} = 2$ A. [Equation 6](#) gives a minimum value of 52 μ F and a maximum ESR of 0.11 Ω . Assuming a 20% tolerance and a 10% bias de-rating, we arrive at a minimum capacitance of 72 μ F. This can be achieved with a bank of 4 $\times$ 22- μ F, 16-V, ceramic capacitors in the 1210 case size. More output capacitance can be used to improve the load transient response. Ceramic capacitors can easily meet the minimum ESR requirements. In some cases an aluminum electrolytic capacitor can be placed in parallel with the ceramics to help build up the required value of capacitance. In general use a capacitor of at least 10 V for output voltages of 3.3 V or less, while a capacitor of 16 V or more should be used for output voltages of 5 V and above.

In practice the output capacitor has the most influence on the transient response and loop phase margin. Load transient testing and Bode plots are the best way to validate any given design and should always be completed before the application goes into production. In addition to the required output capacitance, a small ceramic placed on the output can help to reduce high frequency noise. Small case size ceramic capacitors in the range of 1 nF to 100 nF can be very helpful in reducing spikes on the output caused by inductor and board parasitics.

The maximum value of total output capacitance should be limited to about 10 times the design value, or 1000 µF, whichever is smaller. Large values of output capacitance can adversely affect the start-up behavior of the regulator as well as the loop stability. If values larger than noted here must be used, then a careful study of start-up at full load and loop stability must be performed.

8.2.2.5 Input Capacitor Selection

The ceramic input capacitors provide a low impedance source to the regulator in addition to supplying the ripple current and isolating switching noise from other circuits. A minimum of 10 µF of ceramic capacitance is required on the input of the LMR33630. This must be rated for at least the maximum input voltage that the application requires; preferably twice the maximum input voltage. This capacitance can be increased to help reduce input voltage ripple and/or maintain the input voltage during load transients. In addition a small case size 220-nF ceramic capacitor must be used at the input, as close as possible to the regulator. This provides a high frequency bypass for the control circuits internal to the device. For this example a 10-µF, 50-V, X7R (or better) ceramic capacitors is chosen. The 220 nF must also be rated at 50 V with an X7R dielectric.

Many times it is desirable to use an electrolytic capacitor on the input in parallel with the ceramics. This is especially true if long leads/traces are used to connect the input supply to the regulator. The moderate ESR of this capacitor can help damp any ringing on the input supply caused by the long power leads. The use of this additional capacitor also helps with voltage dips caused by input supplies with unusually high impedance.

Most of the input switching current passes through the ceramic input capacitor(s). The approximate RMS value of this current can be calculated from [Equation 7](#) and should be checked against the manufacturers' maximum ratings.

$$I_{RMS} \cong \frac{I_{OUT}}{2} \quad (7)$$

8.2.2.6 CBOOT

The LMR33630 requires a *boot-strap* capacitor connected between the BOOT pin and the SW pin. This capacitor stores energy that is used to supply the gate drivers for the power MOSFETs. A high-quality ceramic capacitor of 100 nF and at least 6.3 V is required.

8.2.2.7 VCC

The VCC pin is the output of the internal LDO used to supply the control circuits of the regulator. This output requires a 1-µF, 10-V ceramic capacitor connected from VCC to GND for proper operation. In general this output should not be loaded with any external circuitry. However, this output can be used to supply the pull up for the power good function (see [Power-Good Flag Output](#)). A value of 100 kΩ is a good choice in this case. The nominal output voltage on VCC is 5 V; see [Electrical Characteristics](#) for limits.

8.2.2.8 C_{FF} Selection

In some cases a feed-forward capacitor can be used across R_{FBT} to improve the load transient response or improve the loop-phase margin. [Equation 8](#) can be used to estimate the value of C_{FF}. The value found with [Equation 8](#) is a starting point; use lower values to determine if any advantage is gained by the use of a C_{FF} capacitor. The application report [Optimizing Transient Response of Internally Compensated DC-DC Converters with Feed-forward Capacitor](#) is helpful when experimenting with a feed-forward capacitor.

$$C_{FF} < \frac{V_{OUT} \cdot C_{OUT}}{120 \cdot R_{FBT} \cdot \sqrt{\frac{V_{REF}}{V_{OUT}}}} \quad (8)$$

8.2.2.8.1 External UVLO

In some cases an input UVLO level different than that provided internal to the device is needed. This can be accomplished by using the circuit shown in [Figure 17](#) can be used. The input voltage at which the device turns-on is designated V_{ON} ; while the turn-off voltage is V_{OFF} . First a value for R_{ENB} is chosen in the range of 10 k Ω to 100 k Ω and then [Equation 9](#) is used to calculate R_{ENT} and V_{OFF} .

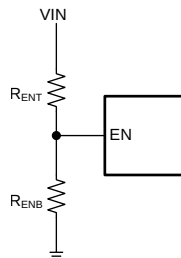


Figure 17. Set-up for External UVLO Application

$$R_{ENT} = \left(\frac{V_{ON}}{V_{EN-H}} - 1 \right) \cdot R_{ENB}$$

$$V_{OFF} = V_{ON} \cdot \left(1 - \frac{V_{EN-HYS}}{V_{EN}} \right)$$

where

- $V_{ON} = V_{IN}$ turnon voltage
- $V_{OFF} = V_{IN}$ turnoff voltage

(9)

8.2.2.9 Maximum Ambient Temperature

As with any power conversion device, the LMR33630 dissipates internal power while operating. The effect of this power dissipation is to raise the internal temperature of the converter above ambient. The internal die temperature (T_J) is a function of the ambient temperature, the power loss and the effective thermal resistance, $R_{\theta JA}$ of the device and PCB combination. The maximum internal die temperature for the LMR33630 must be limited to 125°C. This establishes a limit on the maximum device power dissipation and therefore the load current. [Equation 10](#) shows the relationships between the important parameters. It is easy to see that larger ambient temperatures (T_A) and larger values of $R_{\theta JA}$ reduces the maximum available output current. As stated in [Semiconductor and IC Package Thermal Metrics](#), the values given in [Thermal Information](#) are not valid for design purposes and must not be used to estimate the thermal performance of the application. The values reported in that table were measured under a specific set of conditions that are rarely obtained in an actual application.

$$I_{OUT}|_{MAX} = \frac{(T_J - T_A)}{R_{\theta JA}} \cdot \frac{\eta}{(1 - \eta)} \cdot \frac{1}{V_{OUT}}$$

where

- η = Efficiency

(10)

The effective $R_{\theta JA}$ is a critical parameter and depends on many factors such as power dissipation, air temperature/flow, PCB area, copper heat-sink area, number of thermal vias under the package, and adjacent component placement; to mention just a few. The SOIC (DDA) package utilizes a die attached paddle (DAP) to provide a place to solder down to the PCB heat-sinking copper. This provides a good heat conduction path from the regulator junction to the heat sink and must be properly soldered to the PCB heat sink copper. A typical example of $R_{\theta JA}$ vs copper board area can be found in [Figure 18](#). The copper area given in the graph is for each layer; the top and bottom layers are 2 oz. (70 μ m) copper each. A typical curve of maximum output current vs. ambient temperature is shown in [Figure 19](#). This data was taken with a device/PCB combination giving an $R_{\theta JA}$ of about 30 °C/W. It must be remembered that the data given in these graphs are for illustration purposes only, and the actual performance in any given application depends on all of the factors mentioned above.

The resources below can be used as a guide to optimal thermal PCB design and estimating $R_{\theta JA}$ for a given application environment:

- [Thermal Design by Insight not Hindsight](#)
- [A Guide to Board Layout for Best Thermal Resistance for Exposed Pad Packages](#)
- [Semiconductor and IC Package Thermal Metrics](#)
- [Thermal Design Made Simple with LM43603 and LM43602](#)
- [SLMA002 PowerPAD™ Thermally Enhanced Package](#)
- [PowerPAD™ Made Easy](#)
- [SBVA025 Using New Thermal Metrics](#)

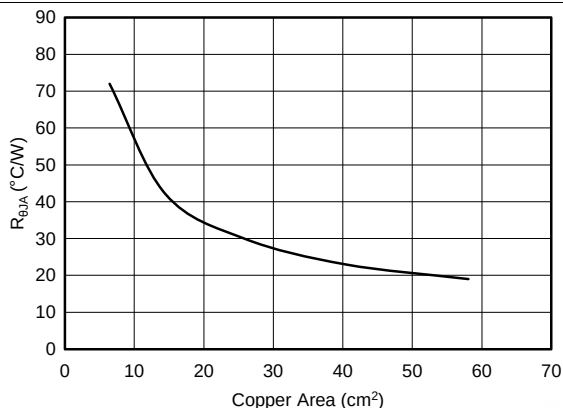


Figure 18. Typical $R_{\theta JA}$ vs Copper Area for a Two-Layer Board and the HSOIC Package

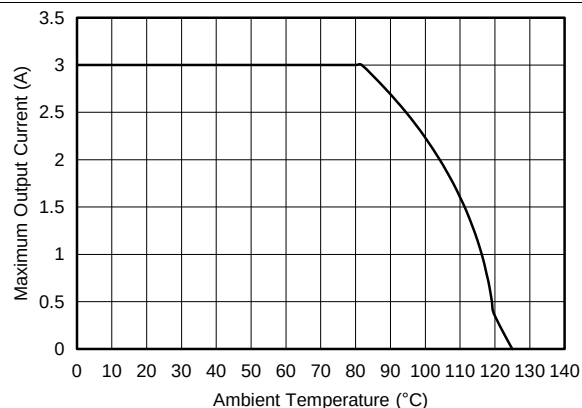


Figure 19. Maximum Output Current vs Ambient Temperature
 $V_{IN} = 12\text{ V}$, $V_{OUT} = 5\text{ V}$, $f_{SW} = 400\text{ kHz}$, $R_{\theta JA} = 30^\circ\text{C/W}$

8.2.3 Application Curves

Unless otherwise specified the following conditions apply: $V_{IN} = 12\text{ V}$, $T_A = 25^\circ\text{C}$. The circuit is shown in [Figure 34](#), with the appropriate BOM from [Table 3](#).

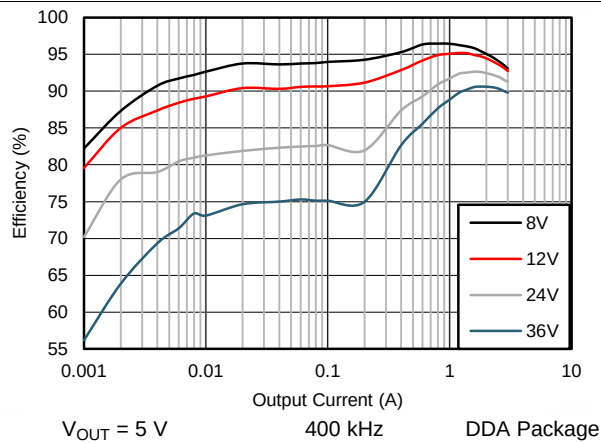


Figure 20. Efficiency

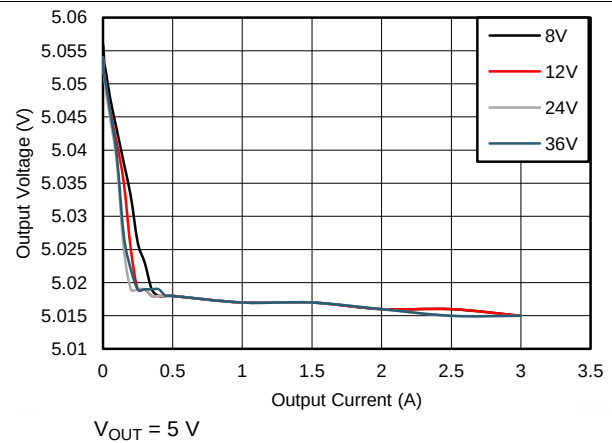


Figure 21. Line and Load Regulation

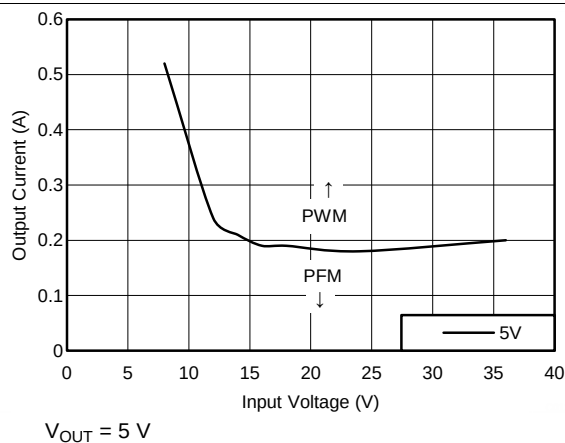


Figure 22. Mode Change Thresholds

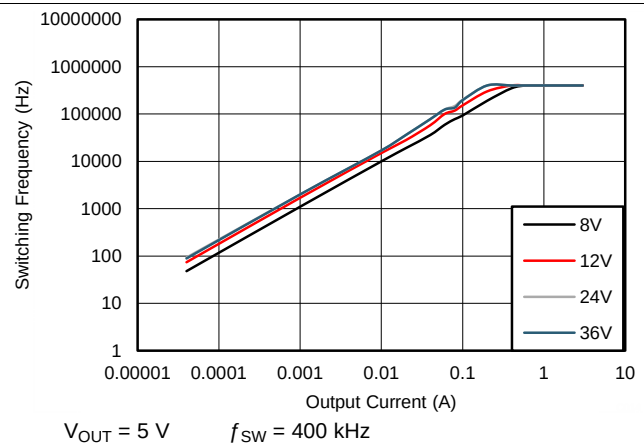


Figure 23. Switching Frequency vs Output Current

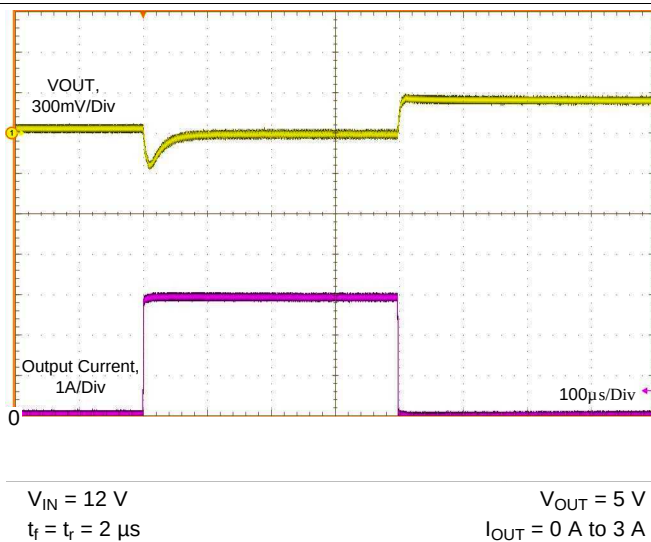


Figure 24. Load Transient

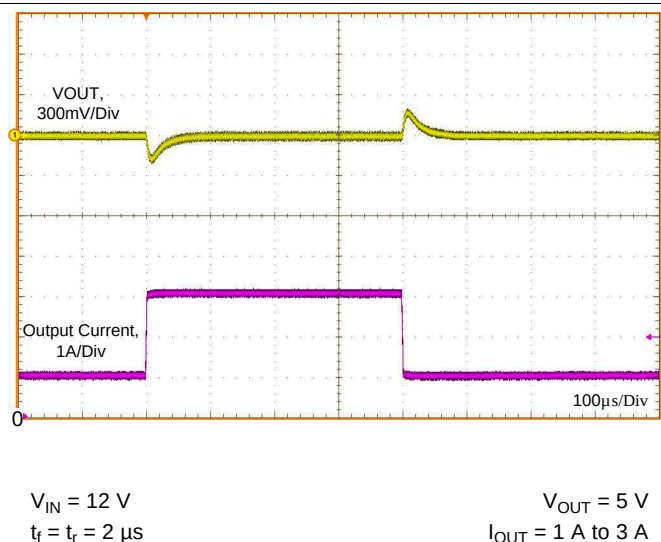


Figure 25. Load Transient

Unless otherwise specified the following conditions apply: $V_{IN} = 12\text{ V}$, $T_A = 25^\circ\text{C}$. The circuit is shown in [Figure 34](#), with the appropriate BOM from [Table 3](#).

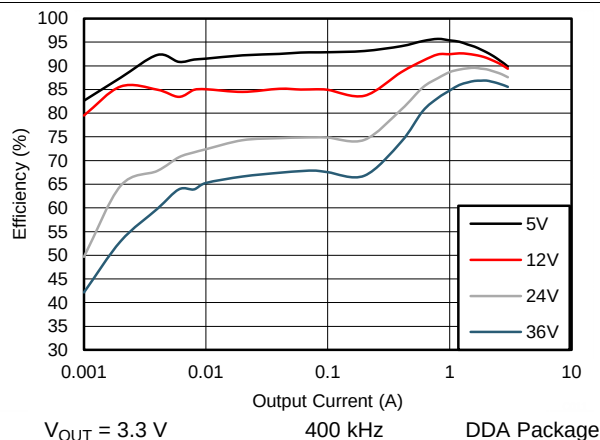


Figure 26. Efficiency

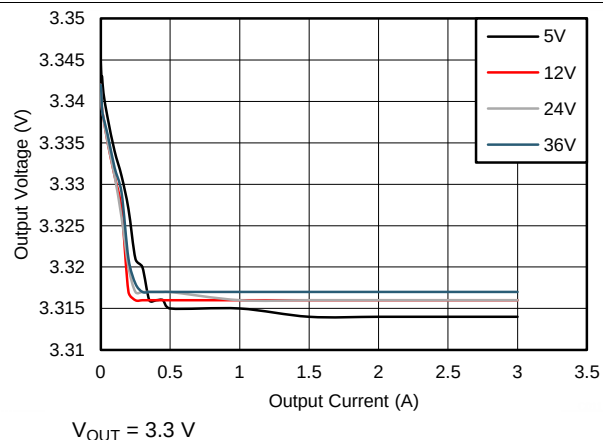


Figure 27. Line and Load Regulation

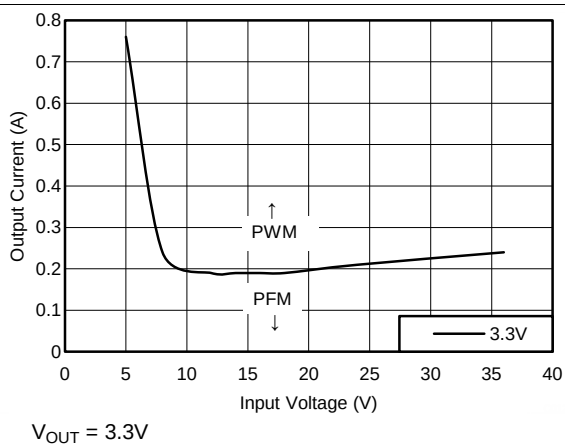


Figure 28. Mode Change Thresholds

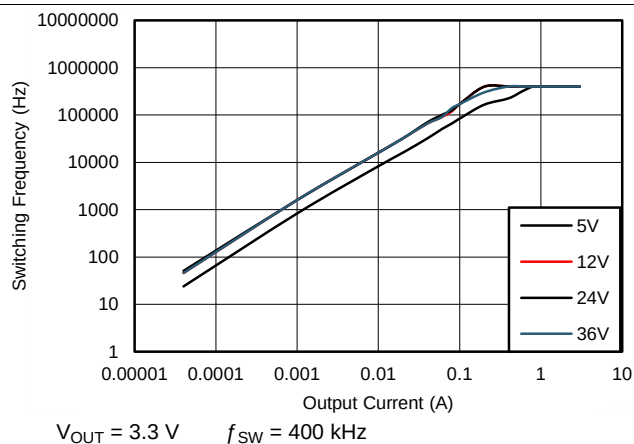


Figure 29. Switching Frequency vs Output Current

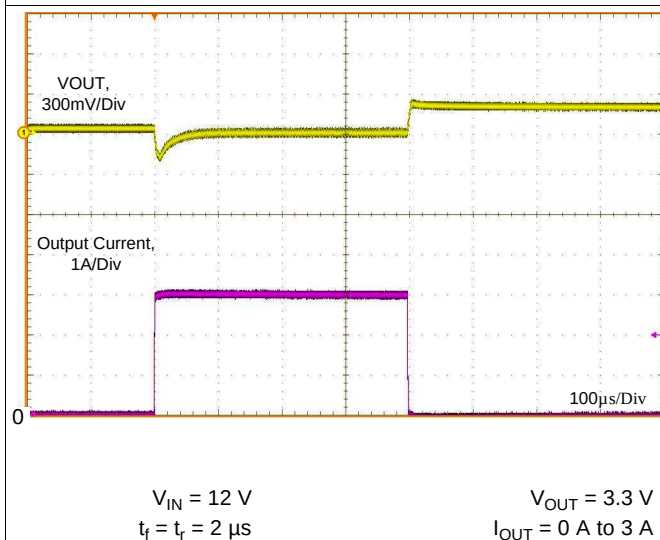


Figure 30. Load Transient

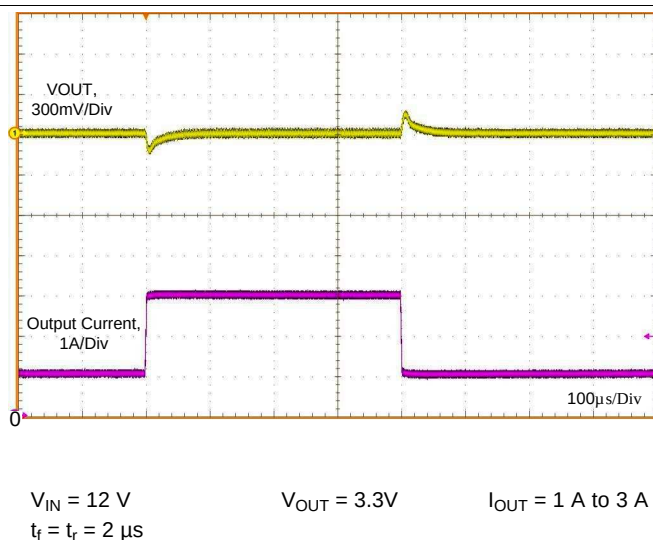
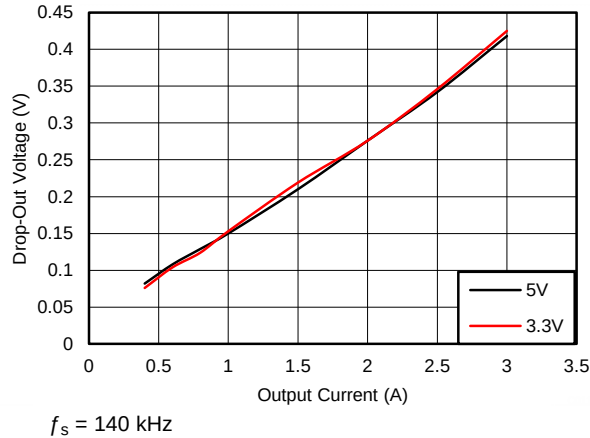
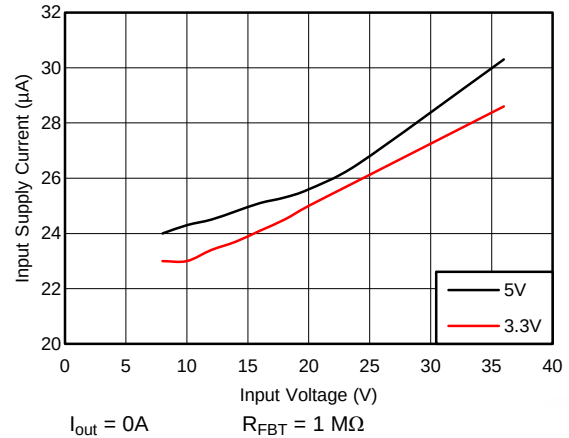
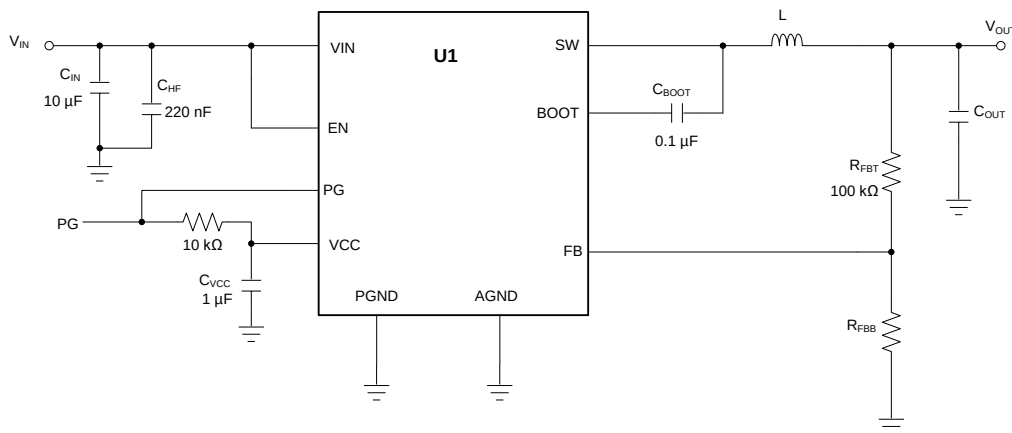


Figure 31. Load Transient

Unless otherwise specified the following conditions apply: $V_{IN} = 12\text{ V}$, $T_A = 25^\circ\text{C}$. The circuit is shown in [Figure 34](#), with the appropriate BOM from [Table 3](#).


Figure 32. Dropout Voltage

Figure 33. Input Supply Current

Figure 34. Circuit for Typical Application Curves
Table 3. BOM for Typical Application Curves

V_{OUT}	FREQUENCY	R_{FBB}	C_{OUT}	L	U1
3.3 V	400 kHz	43.3 kΩ	4 × 22µF	6.8 µH, 14 mΩ	LMR33630ADDA
5 V	400 kHz	24.9 kΩ	4 × 22µF	8 µH, 14 mΩ	LMR33630ADDA

8.3 Do's and Don'ts

- Don't:** Exceed the [Absolute Maximum Ratings](#)
- Don't:** Exceed the [ESD Ratings](#)
- Don't:** Exceed the [Recommended Operating Conditions](#)
- Don't:** Allow the EN input to float.
- Don't:** Allow the output voltage to exceed the input voltage, nor go below ground.
- Don't:** Use the thermal data given in the [Thermal Information](#) table to design your application.
- Do:** Follow all the guidelines and/or suggestions found in this data sheet before committing the design to production. TI application engineers are ready to help critique your design and PCB layout to help make your project a success (see [Community Resources](#)).

9 Power Supply Recommendations

The characteristics of the input supply must be compatible with the [Absolute Maximum Ratings](#) and [Recommended Operating Conditions](#) found in this data sheet. In addition, the input supply must be capable of delivering the required input current to the loaded regulator. The average input current can be estimated with [Equation 11](#), where η is the efficiency.

$$I_{IN} = \frac{V_{OUT} \cdot I_{OUT}}{V_{IN} \cdot \eta} \quad (11)$$

If the regulator is connected to the input supply through long wires or PCB traces, special care is required to achieve good performance. The parasitic inductance and resistance of the input cables can have an adverse effect on the operation of the regulator. The parasitic inductance, in combination with the low-ESR, ceramic input capacitors, can form an underdamped resonant circuit, resulting in overvoltage transients at the input to the regulator. The parasitic resistance can cause the voltage at the VIN pin to dip whenever a load transient is applied to the output. If the application is operating close to the minimum input voltage, this dip may cause the regulator to momentarily shutdown and/or reset. The best way to solve these kind of issues is to reduce the distance from the input supply to the regulator and/or use an aluminum or tantalum input capacitor in parallel with the ceramics. The moderate ESR of these types of capacitors help to damp the input resonant circuit and reduce any overshoots. A value in the range of 20 μ F to 100 μ F is usually sufficient to provide input damping and help to hold the input voltage steady during large load transients.

Sometimes, for other system considerations, an input filter is used in front of the regulator. This can lead to instability, as well as some of the effects mentioned above, unless it is designed carefully. The user guide [AN-2162 Simple Success With Conducted EMI From DCDC Converters](#) provides helpful suggestions when designing an input filter for any switching regulator.

In some cases a transient voltage suppressor (TVS) is used on the input of regulators. One class of this device has a *snap-back* characteristic (thyristor type). The use of a device with this type of characteristic is not recommended. When the TVS fires, the clamping voltage falls to a very low value. If this voltage is less than the output voltage of the regulator, the output capacitors discharge through the device back to the input. This uncontrolled current flow may damage the device.

10 Layout

10.1 Layout Guidelines

The PCB layout of any DC-DC converter is critical to the optimal performance of the design. Bad PCB layout can disrupt the operation of an otherwise good schematic design. Even if the converter regulates correctly, bad PCB layout can mean the difference between a robust design and one that cannot be mass produced. Furthermore, the EMI performance of the regulator is dependent on the PCB layout, to a great extent. In a buck converters the most critical PCB feature is the loop formed by the input capacitor(s) and power ground, as shown in [Figure 35](#). This loop carries large transient currents that can cause large transient voltages when reacting with the trace inductance. These unwanted transients voltages will disrupt the proper operation of the converter. Because of this, the traces in this loop should be wide and short, and the loop area as small as possible to reduce the parasitic inductance. [Figure 36](#) shows a recommended layout for the critical components of the LMR33630.

1. **Place the input capacitor(s) as close as possible to the VIN and GND terminals.** VIN and GND pins are adjacent, simplifying the input capacitor placement.
2. **Place bypass capacitor for VCC close to the VCC pin.** This capacitor must be placed close to the device and routed with short, wide traces to VCC pin and GND.
3. **Use wide traces for the CBOOT capacitor.** Place CBOOT close to the device with short/wide traces to the BOOT and SW pins.
4. **Place the feedback divider as close as possible to the FB pin of the device.** Place R_{FBB} , R_{FBT} , and C_{FF} , if used, physically close to the device. The connections to FB and GND must be short and close to those pins on the device. The connection to V_{OUT} can be somewhat longer. However, this latter trace must not be routed near any noise source (such as the SW node) that can capacitively couple into the feedback path of the regulator.
5. **Use at least one ground plane in one of the middle layers.** This plane acts as a noise shield and also act as a heat dissipation path.
6. **Connect the thermal pad to the ground plane.** The SOIC package has a thermal pad (PAD) connection that must be soldered down to the PCB ground plane. This pad acts as a heat-sink connection and an electrical ground connection for the regulator. The integrity of this solder connection has a direct bearing on the total effective $R_{\theta JA}$ of the application.
7. **Provide wide paths for VIN, VOUT, and GND.** Making these paths as wide and direct as possible reduces any voltage drops on the input or output paths of the converter and maximizes efficiency.
8. **Provide enough PCB area for proper heat-sinking.** As stated in the [Maximum Ambient Temperature](#) section, enough copper area must be used to ensure a low $R_{\theta JA}$, commensurate with the maximum load current and ambient temperature. The top and bottom PCB layers should be made with two ounce copper; and no less than one ounce. With the SOIC package, use an array of heat-sinking vias to connect the thermal pad (PAD) to the ground plane on the bottom PCB layer. If the PCB design uses multiple copper layers (recommended), these thermal vias can also be connected to the inner layer heat-spreading ground planes.
9. **Keep switch area small.** Keep the copper area connecting the SW pin to the inductor as short and wide as possible. At the same time the total area of this node should be minimized to help reduce radiated EMI.

See the following PCB layout resources for additional important guidelines:

- [Layout Guidelines for Switching Power Supplies](#)
- [Simple Switcher PCB Layout Guidelines](#)
- [Construction Your Power Supply- Layout Considerations](#)
- [Low Radiated EMI Layout Made Simple with LM4360x and LM4600x](#)

Layout Guidelines (continued)

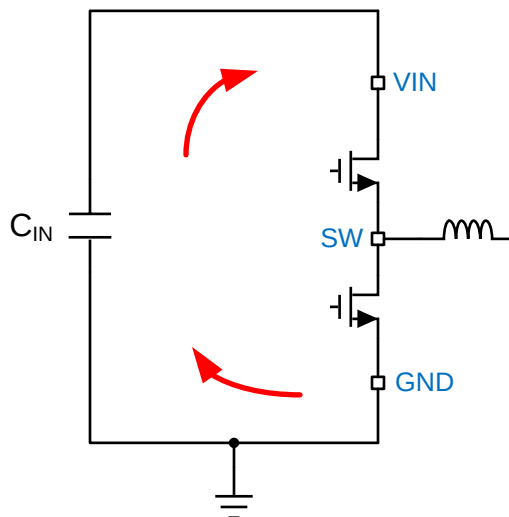


Figure 35. Current Loops with Fast Edges

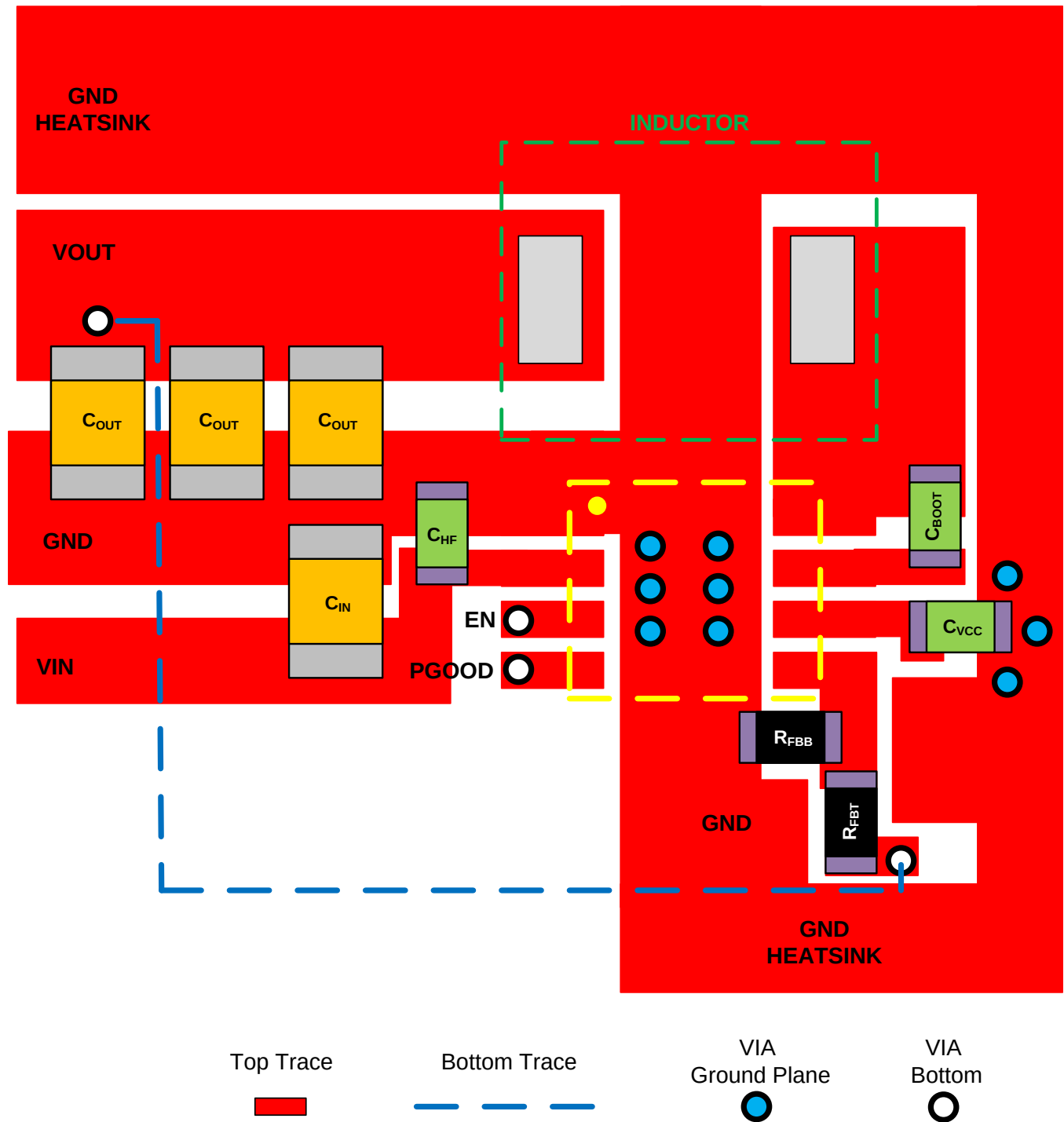
10.1.1 Ground and Thermal Considerations

As mentioned above, TI recommends using one of the middle layers as a solid ground plane. A ground plane provides shielding for sensitive circuits and traces. It also provides a quiet reference potential for the control circuitry. The AGND and PGND pins should be connected to the ground planes using vias next to the bypass capacitors. PGND pins are connected directly to the source of the low side MOSFET switch, and also connected directly to the grounds of the input and output capacitors. The PGND net contains noise at the switching frequency and may bounce due to load variations. The PGND trace, as well as the VIN and SW traces, must be constrained to one side of the ground planes. The other side of the ground plane contains much less noise and must be used for sensitive routes.

TI recommends providing adequate device heat sinking by utilizing the thermal pad (PAD) of the device as the primary thermal path. Use a minimum 4 × 3 array of 10 mil thermal vias to connect the PAD to the system ground plane heat sink. The vias must be evenly distributed under the PAD. Use as much copper as possible, for system ground plane, on the top and bottom layers for the best heat dissipation. Use a four-layer board with the copper thickness for the four layers, starting from the top as: 2 oz / 1 oz / 1 oz / 2 oz. A four-layer board with enough copper thickness, and proper layout, provides low current conduction impedance, proper shielding and lower thermal resistance.

10.2 Layout Example

ADVANCE INFORMATION



11 Device and Documentation Support

11.1 Documentation Support

11.1.1 Related Documentation

For related documentation see the following:

- [Thermal Design by Insight not Hindsight](#)
- [A Guide to Board Layout for Best Thermal Resistance for Exposed Pad Packages](#)
- [Semiconductor and IC Package Thermal Metrics](#)
- [Thermal Design Made Simple with LM43603 and LM43602](#)
- [SLMA002 PowerPAD™ Thermally Enhanced Package](#)
- [PowerPAD™ Made Easy](#)
- [SBVA025 Using New Thermal Metrics](#)
- [Layout Guidelines for Switching Power Supplies](#)
- [Simple Switcher PCB Layout Guidelines](#)
- [Construction Your Power Supply- Layout Considerations](#)
- [Low Radiated EMI Layout Made Simple with LM4360x and LM4600x](#)

11.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

11.3 Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

TI E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

11.4 Trademarks

PowerPAD, E2E are trademarks of Texas Instruments.

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11.5 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

11.6 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

12 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
PLMR33630ADDA	ACTIVE	SO PowerPAD	DDA	8	75	TBD	Call TI	Call TI	-40 to 125		Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

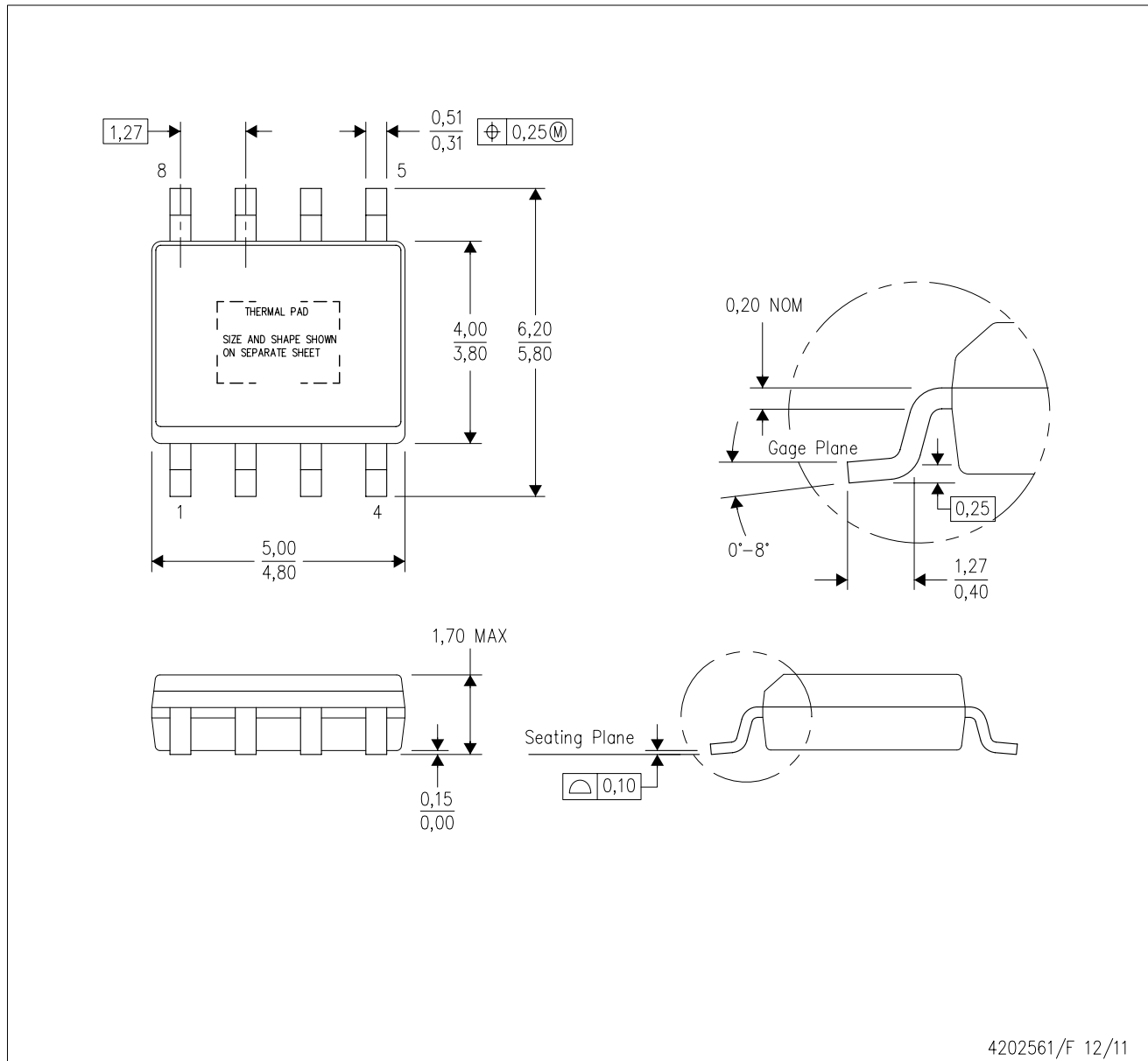
(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

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DDA (R-PDSO-G8)

PowerPAD™ PLASTIC SMALL-OUTLINE



- NOTES:
- All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5-1994.
 - This drawing is subject to change without notice.
 - Body dimensions do not include mold flash or protrusion not to exceed 0,15.
 - This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <<http://www.ti.com>>.
 - See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
 - This package complies to JEDEC MS-012 variation BA

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